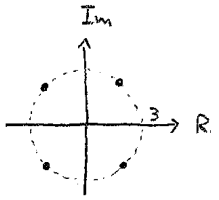
複素解析

1. $z^4 + 81 = r^4 e^{i4\theta} + 81 = 0 \qquad r = 3 \qquad 4\theta = (2n+1)\pi \qquad n = 0, 1, 2, \cdots$



2. $$\prod_{k=1}^{m-1} \sin\frac{k\pi}{m} = \prod_{k=1}^{m-1} \frac{e^{i\frac{k\pi}{m}} - e^{-i\frac{k\pi}{m}}}{2i} = \frac{e^{i\sum_{k=1}^{m-1}\frac{k\pi}{m}}}{(2i)^{m-1}} \prod_{k=1}^{m-1} \left(1 - e^{-i\frac{2\pi}{m}k}\right)$$

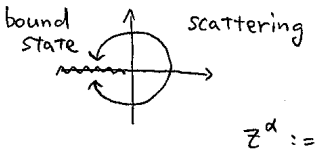
$$= \frac{e^{i\frac{\pi}{2}(m-1)}}{(2i)^{m-1}} \prod_{k=1}^{m-1} (1 - z_k) = \frac{1}{2^{m-1}} \cdot \frac{\prod_{k=0}^{m-1}(1 - z_k)}{1 - z_0} = \frac{1}{2^{m-1}} \cdot \frac{1-x^m}{1-x}\Big|_{x=1}$$

$$= \frac{1}{2^{m-1}} (1 + \cdots + x^{m-1})\Big|_{x=1} = \frac{m}{2^{m-1}}$$

3. (a) $re^{i\theta} = e^w = e^u \cdot e^{iv} \qquad u = \ln r \qquad v = \theta + 2\pi n$ 対価性

(b) $\ln(1-i) = \ln\sqrt{2} + i\left(-\frac{\pi}{4} + 2\pi n\right)$

主値 $= \ln\sqrt{2} - \frac{i\pi}{4}$



4. (a) $z = \sin w = \frac{e^{iw} - e^{-iw}}{2i}$

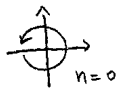
$$(e^{iw})^2 - 2iz(e^{iw}) - 1 = 0 \qquad e^{iw} = iz \pm \sqrt{1-z^2}$$

↖ $\sin^{-1} 0 = 0$ より plus

$$z = \tanh w = \frac{e^w - e^{-w}}{e^w + e^{-w}}$$

$$e^{2w} = \frac{1+z}{1-z}$$

(b) $i^i = e^{i\ln i} = e^{i\left(i\frac{\pi}{2} + i2\pi n\right)} = e^{-\frac{\pi}{2}} \left(e^{-2\pi}\right)^n$



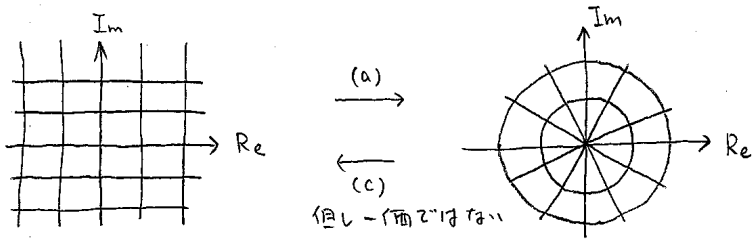
5. (a) $(1+i)^i = e^{i\ln(1+i)} = e^{i\left(\ln\sqrt{2} + i\left(\frac{\pi}{4} + 2\pi n\right)\right)} = e^{-\left(\frac{\pi}{4} + 2\pi n\right)} e^{i\ln\sqrt{2}}$

(b) $1^{\sqrt{2}} = e^{\sqrt{2}\ln 1} = e^{\sqrt{2}(i2\pi n)} = e^{i2\sqrt{2}\pi m}$

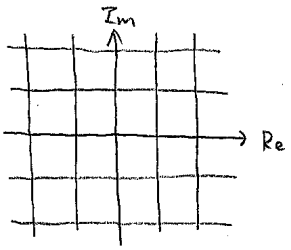
(c) $(1-i)^{1+i} = e^{(1+i)\ln(1-i)} = e^{(1+i)\left(\ln\sqrt{2} + i\left(-\frac{\pi}{4} + 2\pi n\right)\right)}$

$$= e^{\ln\sqrt{2} - \left(-\frac{\pi}{4} + 2\pi n\right)} e^{i\left(\ln\sqrt{2} - \frac{\pi}{4} + 2\pi n\right)}$$

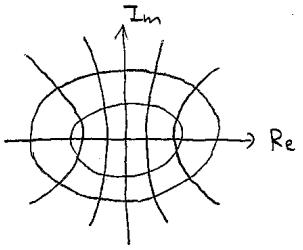
6.



(b) $\cos z = \frac{e^{iz}+e^{-iz}}{2} = \frac{e^{ix}e^{-y}+e^{-ix}e^{y}}{2} = \cos x \cosh y + i \sin x \sinh y$



(b) →



$\frac{u^2}{\cosh^2 y} + \frac{v^2}{\sinh^2 y} = 1$

$\frac{u^2}{\cos^2 x} - \frac{v^2}{\sin^2 x} = 1$

(d) $z^3 = x^3 - 3xy^2 + i(3x^2y - y^3)$

7.

(a) $z = x + iy \quad (x, y \in \mathbb{R})$

$f(z) = u + iv$

$$\begin{cases} \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \\ \frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y} \end{cases} \quad \cdots (*)$$ Cauchy-Riemann condition

$\longrightarrow \Delta u = \Delta v = 0 \qquad \Delta = \partial_x^2 + \partial_y^2$

$df = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy \qquad \begin{cases} dx = \frac{1}{2}(dz + d\bar{z}) \\ dy = \frac{1}{2i}(dz - d\bar{z}) \end{cases}$

$= \frac{1}{2}\left(\frac{\partial f}{\partial x} + \frac{1}{i}\frac{\partial f}{\partial y}\right) dz + \frac{1}{2}\left(\frac{\partial f}{\partial x} - \frac{1}{i}\frac{\partial f}{\partial y}\right) d\bar{z}$

$df(z) = (P(x,y) + iQ(x,y))(dx + i\,dy)$

$P(x,y) = \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$

$Q(x,y) = \frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y}$

(b) $\frac{\partial}{\partial \bar{z}} = \frac{\partial x}{\partial \bar{z}}\frac{\partial}{\partial x} + \frac{\partial y}{\partial \bar{z}}\frac{\partial}{\partial y} =$

8. (a) $u = 2x(1-y)$

$$dv = \frac{\partial v}{\partial x}dx + \frac{\partial v}{\partial y}dy = -\frac{\partial u}{\partial y}dx + \frac{\partial u}{\partial x}dy \qquad \text{Cauchy-Riemann condition}$$

$$= 2x\,dx + 2(1-y)\,dy$$

$$v = x^2 - (1-y)^2 + \text{const}$$

$$f(z) = u + iv = 2x(1-y) + ix^2 - i(1-y)^2 + i(\text{const}) = z + iz^2 + i(\text{const})$$

(b) $$\frac{\partial^2\psi}{\partial x^2} = \frac{\partial}{\partial x}\left(\frac{2(x-a)}{(x-a)^2+(y-b)^2}\right) = 2\,\frac{-(x-a)^2+(y-b)^2}{((x-a)^2+(y-b)^2)^2} \rightarrow \Delta\psi = 0 \quad \text{harmonic function}$$

$$d\phi = \frac{\partial\phi}{\partial x}dx + \frac{\partial\phi}{\partial y}dy = -\frac{\partial\psi}{\partial y}dx + \frac{\partial\psi}{\partial x}dy$$

$$= -\frac{2(y-b)}{(x-a)^2+(y-b)^2}dx + \frac{2(x-a)}{(x-a)^2+(y-b)^2}dy$$

$$\psi = \ln\{(x-a)+i(y-b)\}\{(x-a)-i(y-b)\} \qquad z = x+iy$$

$$= \ln\{(z-\alpha)(\bar{z}-\bar{\alpha})\} = \ln|z-\alpha|^2 \qquad \alpha = a+ib$$

$$\phi = -i\ln(\bar{z}-\bar{\alpha}) + c.c$$

$$= -i\ln(\bar{z}-\bar{\alpha}) + i\ln(z-\alpha) = i\ln\frac{z-\alpha}{\bar{z}-\bar{\alpha}} \qquad f = \phi + i\psi$$

9. $$\frac{R^2-r^2}{R^2-2Rr\cos(\theta-\varphi)+r^2} = \frac{|Z|^2-|z|^2}{|Z-z|^2} = \frac{Z\bar{Z}-z\bar{z}}{(Z-z)(\bar{Z}-\bar{z})} = \frac{1}{2}\left(\frac{Z+z}{Z-z} + \frac{\bar{Z}+\bar{z}}{\bar{Z}-\bar{z}}\right)$$

$$= \mathrm{Re}\,\frac{Z+z}{Z-z}$$

10.

(a) $$\frac{z^2-2z}{(z+1)^2(z^2+4)}$$

$$\mathrm{Res}(2i) = \left.\frac{z^2-2z}{(z+1)^2(z+2i)}\right|_{z=2i} = \frac{1-i}{3-4i} = \frac{7+i}{25}$$

$$\mathrm{Res}(-2i) = \left.\frac{z^2-2z}{(z+1)^2(z-2i)}\right|_{z=-2i} = \frac{1+i}{3+4i} = \frac{7-i}{25}$$

$$\mathrm{Res}(-1) = \left.\frac{d}{dz}\,\frac{z^2-2z}{z^2+4}\right|_{z=-1} = \left.\frac{(2z-2)(z^2+4)-(z^2-2z)\cdot 2z}{(z^2+4)^2}\right|_{z=-1} = -\frac{14}{25}$$

$f(z) = \frac{P(z)}{Q(z)}$ $\quad \text{Res}(f, a) = (z-a)\frac{P(z)}{Q(z)}\Big|_{z=a} = \frac{z-a}{Q(z)-Q(a)}P(z)\Big|_{z=a} = \frac{P(a)}{Q'(a)}$

(b) $\text{sech}(z) = \frac{1}{\text{Cosh}(z)}$ $\quad \text{Cosh}(a) = 0 \quad a = i(n+\frac{1}{2})\pi$

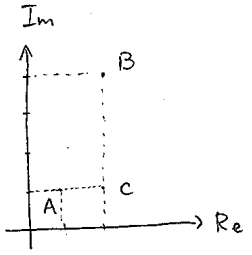
$$\text{Res}(f, a) = \frac{z-a}{\text{Cosh}(z)}\Big|_{z=a} = \frac{z-a}{\text{Cosh}(z)-\text{Cosh}(a)}\Big|_{z=a} = \left[\text{Cosh}'(z)\right]^{-1}_{z=a} = \frac{2}{e^a - e^{-a}}$$

$$= \frac{2}{i(-1)^n - \frac{1}{i(-1)^n}} = -(-1)^n i$$

(c) $\cot(z) = \frac{1}{\tan(z)} = \frac{\text{Cos}(z)}{\sin(z)} \doteq \frac{1-\frac{z^2}{2}}{z}$ $\quad \text{Res}(f, 0) = 1$

(d) $e^z \text{cosec}^2 z = \frac{e^z}{\sin^2 z} \doteq \frac{1+z}{z^2}$ $\quad \text{Res}(f, 0) = 1$

11. $I = \int_A^B z^2 dz$



(a) $z = t + it^2 \quad dz = (1+2it)dt$

$$I = \int_1^2 (t+it^2)^2(1+2it)dt$$

$$= \int_1^2 \left\{(-5t^4+t^2) + i(-2t^5+4t^3)\right\}dt = \left[-t^5+\frac{t^3}{3}\right]_1^2 + i\left[-\frac{t^6}{3}+t^4\right]_1^2 = -\frac{86}{3} - 6i$$

(b) $z = t + i(3t-2) \quad dz = (1+3i)dt$

$$I = \int_1^2 ((1+3i)t-2i)^2(1+3i)dt = \frac{1}{3}\left[((1+3i)t-2i)^3\right]_1^2 = -\frac{86}{3} - 6i$$

(c) $I = \int_A^C z^2 dz + \int_C^B z^2 dz = \int_1^2 (t+i)^2 dt + \int_1^4 (2+it)^2 i\,dt$

$z = t+i \quad z = 2+it$

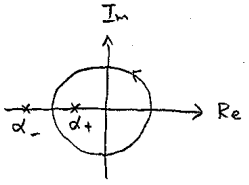
$$= \frac{1}{3}\left[(t+i)^3\right]_1^2 + \frac{1}{3}\left[(2+it)^3\right]_1^4 = -\frac{86}{3} - 6i$$

(d) $I = \int_A^B z^2 dz = \left[\frac{z^3}{3}\right]_A^B = \frac{(2+4i)^3 - (1+i)^3}{3} = -\frac{86}{3} - 6i$

12.

(a) $I = \int_0^{2\pi} \frac{d\theta}{(a+b\cos\theta)^3} = \int_C \frac{dz/iz}{\left(a+\frac{b}{2}\left(z+\frac{1}{z}\right)\right)^3} = \int_C \frac{-8iz^2\,dz}{(bz^2+2az+b)^3}$

3次の pole $\alpha_\pm = \frac{-a\pm\sqrt{a^2-b^2}}{b}$ $\qquad \alpha_- < -1 < \alpha_+ < 0$



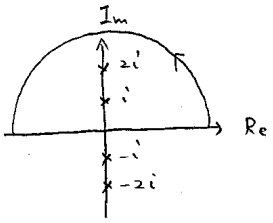
$$\mathrm{Res}(f,\alpha_+) = \frac{1}{2!}\frac{d^2}{dz^2}\frac{-8iz^2}{b^3(z-\alpha_-)^3}\Big|_{z=\alpha_+}$$

$$= -\frac{4i}{b^3}\left(2(z-\alpha_-)^{-3} - 12z(z-\alpha_-)^{-4} + 12z^2(z-\alpha_-)^{-5}\right)_{z=\alpha_+}$$

$$= -\frac{8i\left((\alpha_+-\alpha_-)^2+6\alpha_+\alpha_-\right)}{b^3(\alpha_+-\alpha_-)^5} = -i\,\frac{2a^2+b^2}{2(a^2-b^2)^{5/2}}$$

$$I = 2\pi i\,\mathrm{Res}(f,\alpha_+) = 2\pi\,\frac{2a^2+b^2}{2(a^2-b^2)^{5/2}}$$

(b) $I = \int_0^\infty \frac{dx}{(x^2+1)^2(x^2+4)} = \frac{1}{2}\int_{-\infty}^\infty \frac{dx}{(x^2+1)^2(x^2+4)}$

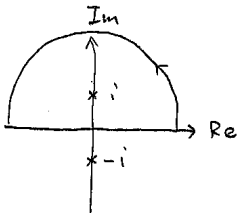


$$\mathrm{Res}(f,2i) = \frac{1}{((2i)^2+1)^2(2i+2i)} = \frac{1}{36i}$$

$$\mathrm{Res}(f,i) = \frac{d}{dx}\frac{1}{(x+i)^2(x^2+4)}\Big|_{x=i} = -2(x+i)^{-3}(x^2+4)^{-1} - (x+i)^{-2}(x^2+4)^{-2}\cdot 2x\Big|_{x=i} = \frac{1}{36i}$$

$$I = \frac{1}{2}\left(2\pi i\,\mathrm{Res}(f,2i) + 2\pi i\,\mathrm{Res}(f,i)\right) = \frac{\pi}{18}$$

(c) $I = \int_{-\infty}^\infty \frac{\cos x}{x^2+1}\,dx = \int_{-\infty}^\infty \frac{e^{ix}}{x^2+1}\,dx = 2\pi i\,\mathrm{Res}(f,i) = \frac{\pi}{e}$



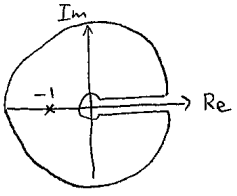
$$\mathrm{Res}(f,i) = \frac{e^{-1}}{i+i}$$

$$\left|\int \frac{e^{ix}}{x^2+1}\,dx\right| \le \int \left|\frac{e^{iR\cos\theta}}{R^2e^{i2\theta}+1}\right|\left|e^{-R\sin\theta}\right|\left|Rie^{i\theta}\right|d\theta \le \frac{e^{-R\sin\theta}}{R}\cdot\pi \to 0 \qquad \begin{pmatrix} R\to\infty \\ \sin\theta>0 \end{pmatrix}$$

上半面

$x = Re^{i\theta}$

(d) $\int_C \frac{(\ln z)^2}{(1+z)^3}\,dz = \int_0^\infty \frac{(\ln x)^2}{(1+x)^3}\,dx + \int_\infty^0 \frac{(\ln x + 2\pi i)^2}{(1+x)^3}\,dx$



$$= -4\pi i\,I + 4\pi^2 \underbrace{\int_0^\infty \frac{dx}{(1+x)^3}}_{\frac{1}{2}}$$

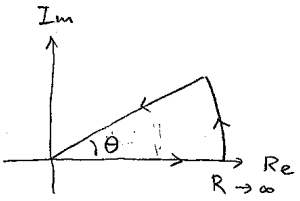
$$\mathrm{Res}(f\ln z,-1) = \frac{1}{2!}\frac{d^2}{dz^2}(\ln z)^2\Big|_{z=-1} = \frac{1-\ln(-1)}{(-1)^2} = 1-i\pi \quad\to\quad I = -\frac{1}{2}$$

$$\left|\int \frac{(\ln z)^2}{(1+z)^3}\,dz\right| \le \int \left|\frac{(\ln R+i\theta)^2}{(1+Re^{i\theta})^3}\right|\left|Rie^{i\theta}\right|d\theta \le \frac{(\ln R)^2}{R^2}\cdot 2\pi \to 0 \quad (R\to\infty)$$

$$\le \int \left|\frac{(\ln\varepsilon+i\theta)^2}{(1+\varepsilon e^{i\theta})^3}\right|\left|\varepsilon ie^{i\theta}\right|d\theta \le (\ln\varepsilon)^2\cdot\varepsilon\cdot 2\pi \to 0 \quad (\varepsilon\to 0)$$

13.

$f(z) = z^{n-1} e^{-z}$ pole なし



$$0 = \int_C f(z)\,dz = \int_0^R x^{n-1} e^{-x}\,dx \longrightarrow \Gamma(n) \quad (R\to\infty)$$

$$+ \int_{\phi=0}^{\phi=\theta} (Re^{i\phi})^{n-1} e^{-R\cos\phi} e^{-iR\sin\phi}\, Rie^{i\phi}\,d\phi \quad \leq \int R^n e^{-R\cos\phi}\,d\phi \longrightarrow 0 \quad (R\to\infty)$$

$$+ \int_R^0 (re^{i\theta})^{n-1} e^{-r\cos\theta} e^{-ir\sin\theta} e^{i\theta}\,dr$$

$\cos\theta = kp$
$\sin\theta = kq$

$$= \Gamma(n) - e^{in\theta} \int_0^\infty r^{n-1} e^{-kpr} e^{-ikqr}\,dr$$

$$= \Gamma(n) - e^{in\theta} \frac{1}{k^n} \int_0^\infty x^{n-1} e^{-px} e^{-iqx}\,dx \qquad kr = x$$

共役とる $\quad \Gamma(n) = \dfrac{e^{-in\theta}}{k^n} \displaystyle\int_0^\infty x^{n-1} e^{-px} e^{iqx}\,dx$

$$\int_0^\infty x^{n-1} e^{-px} \cos(qx)\,dx = \Gamma(n)\,\mathrm{Re}\,(ke^{i\theta})^n = \Gamma(n)\,\frac{\mathrm{Re}\,(p+iq)^n}{(p^2+q^2)^n}$$

$$\int_0^\infty x^{n-1} e^{-px} \sin(qx)\,dx = \Gamma(n)\,\mathrm{Im}\,(ke^{i\theta})^n = \Gamma(n)\,\frac{\mathrm{Im}\,(p+iq)^n}{(p^2+q^2)^n}$$

14.

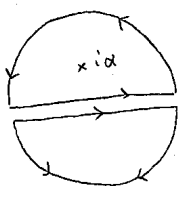
(a) $\lim_{|z|\to\infty} f(z) = 0 \longrightarrow \left| \int_{\text{上半円}} f(z) e^{iz}\,dz \right| \leq |f(R)| \cdot e^{-R} \cdot R \cdot \pi \longrightarrow 0 \quad (R\to\infty)$

analytic だから pole なし $\quad \displaystyle\int_{-\infty}^\infty f(x) e^{ix}\,dx = 0$

$$\left| \int_{\text{上半面}} f(z) e^{-itz}\,dz \right| \leq |f(R)| \cdot e^{tR} \cdot R \cdot \pi \longrightarrow 0 \quad (R\to\infty,\ t<0)$$

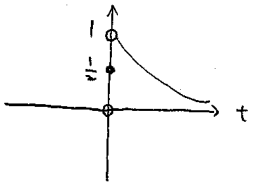
より $\quad \displaystyle\int_{-\infty}^\infty f(s) e^{-its}\,ds = 0$

(b) $e^{itz} = e^{itR\cos\theta} e^{-tR\sin\theta}$ より $\quad t>0$ のとき 上半面
$t<0$ 〃 下半面



$$\theta_\alpha(t) = \frac{1}{2\pi i} \int_{-\infty}^\infty \frac{e^{its}}{s-i\alpha}\,ds = \begin{cases} \mathrm{Res}(i\alpha) = e^{-\alpha t} & (t>0) \\ 0 & (t<0) \end{cases}$$

$$\theta_\alpha(0) = \frac{1}{2\pi i} \int_{-\infty}^\infty \frac{s+i\alpha}{s^2+\alpha^2}\,ds = \frac{\alpha}{2\pi} \int_{-\infty}^\infty \frac{ds}{s^2+\alpha^2} = i\alpha\,\mathrm{Res}(i\alpha) = \frac{1}{2}$$



$$\int_{-\infty}^\infty \theta'(x) f(x)\,dx = \theta(x) f(x)\Big|_{-\infty}^\infty - \int_{-\infty}^\infty \theta(x) f'(x)\,dx = f(\infty) - f(x)\Big|_0^\infty = f(0)$$

15.

$$P\frac{1}{x} = \frac{1}{2}\left\{\frac{1}{x-i\varepsilon} + \frac{1}{x+i\varepsilon}\right\}$$

$$P\int_{-\infty}^{\infty}\frac{f(E')}{E'-E}dE' = \frac{1}{2}\left\{\int_{-\infty}^{\infty}\frac{f(E')}{E'-(E+i\varepsilon)}dE' + \int_{-\infty}^{\infty}\frac{f(E')}{E'-(E-i\varepsilon)}dE'\right\}$$

$$= \frac{1}{2}\left\{0 + 2\pi i\,\mathrm{Res}\left(\frac{f(E')}{E'-E}, E\right)\right\} = \pi i f(E)$$

16.

$g(E) = \dfrac{\tilde{f}(E)}{(E-E_0)^n}$ について $|g(z)|_{|z|\to\infty} \to 0$ であれば 15. より O.K.

$$|g(z)| \leq \frac{|f(z)|}{(z-E_0)^n} + \sum_{k=0}^{n-1}\frac{1}{k!}|f^{(k)}(E_0)|(z-E_0)^{k-n}$$

$$\leq \left|\frac{f(z)}{z^n}\right| + \sum_{k=0}^{n-1}\frac{1}{k!}\left|\frac{f^{(k)}(E_0)}{z^{n-k}}\right| \to 0$$

17. $|\tilde{z}| = \dfrac{R_0^2}{r} > R_0$ $\quad \zeta = R_0 e^{i\theta}$

$$\frac{d\zeta}{\zeta - z} = \frac{\bar{\zeta}-\bar{z}}{|\zeta - z|^2}d\zeta = \frac{R_0 e^{-i\theta} - re^{-i\phi}}{R_0^2 - 2R_0 r\cos(\theta-\phi) + r^2} iR_0 e^{i\theta}d\theta$$

$$\frac{d\zeta}{\zeta - \tilde{z}} = \frac{d\zeta}{\zeta - \left(\frac{R_0}{r}\right)^2 z} = \frac{R_0 e^{-i\theta} - \frac{R_0^2}{r}e^{-i\phi}}{R_0^2 - 2R_0\frac{R_0^2}{r}\cos(\theta-\phi) + \left(\frac{R_0^2}{r}\right)^2} iR_0 e^{i\theta}d\theta$$

$$= \frac{\frac{r^2}{R_0}e^{-i\theta} - re^{-i\phi}}{r^2 - 2R_0 r\cos(\theta-\phi) + R^2} iR_0 e^{i\theta}d\theta$$

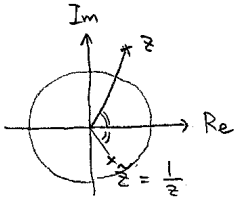
$$f(z) = \frac{1}{2\pi i}\int\frac{f(\zeta)}{\zeta - z}d\zeta - \underbrace{\frac{1}{2\pi i}\int\frac{f(\zeta)}{\zeta - \tilde{z}}d\zeta}_{0}$$

$$= \frac{1}{2\pi}\int\frac{(R_0^2 - r^2)f(\zeta)}{R_0^2 - 2R_0 r\cos(\theta-\phi) + r^2}d\theta$$

18.

(a) $\dfrac{\alpha z+\beta}{\gamma z+\delta} = \dfrac{\alpha}{\gamma} + \dfrac{\beta\gamma-\alpha\delta}{\gamma^2}\cdot\dfrac{1}{z+\frac{\delta}{\gamma}}$

平行移動　回転・拡大　拡大・反転



中心 $-\frac{\delta}{\gamma}$, 半径 R の円 $\longrightarrow$ 中心 $\frac{\alpha}{\gamma}+\left|\frac{\beta\gamma-\alpha\delta}{\gamma^2}\right|\cdot\left(-\frac{\delta}{\gamma}\right)$, 半径 $\frac{1}{R}\left|\frac{\beta\gamma-\alpha\delta}{\gamma^2}\right|$ の円

(b) $(z, z_2 z_3) \xrightarrow{f} (0, \infty, 1)$

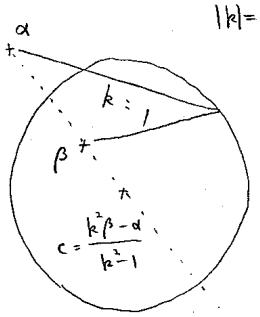
$$f(z) = \frac{z-z_1}{z-z_2} \Big/ \frac{z_3-z_1}{z_3-z_2}$$

$(\omega, \omega_2 \omega_3) \xrightarrow{g} (0, \infty, 1)$

$$g(\omega) = \frac{\omega-\omega_1}{\omega-\omega_2} \Big/ \frac{\omega_3-\omega_1}{\omega_3-\omega_2} \qquad \omega = \frac{\omega_2(\omega_3-\omega_2)\,g(\omega) - \omega_1(\omega_3-\omega_1)}{(\omega_3-\omega_2)\,g(\omega) - (\omega_3-\omega_1)}$$

$g^{-1}\circ f$

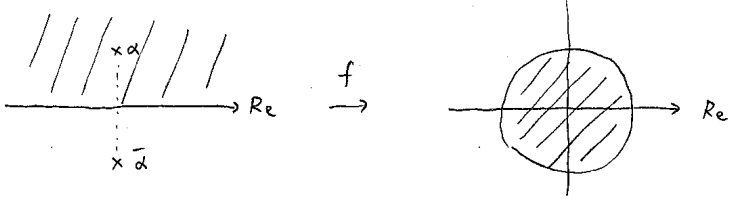
19. (a) $\underbrace{1 = |f(z)|}_{\text{単位円}} = |k|\,\dfrac{|z-\beta|}{|z-\alpha|}$



$|k|=1$　垂直二等分線

$|z-\alpha| = |z-\beta|$

(b) $\omega = f(z) = e^{i\theta}\,\dfrac{z-\alpha}{z-\bar{\alpha}}$　$\mathrm{Im}(\alpha) > 0$



20. $\tilde{\omega} = f(\bar{z}) = e^{i\theta}\,\dfrac{\bar{z}-\alpha}{\bar{z}-\bar{\alpha}} = e^{i\theta}\,\dfrac{\overline{z-\bar{\alpha}}}{\overline{z-\alpha}} = e^{i\theta}\,\overline{\left(\dfrac{z-\bar{\alpha}}{z-\alpha}\right)} = \overline{\left(\dfrac{1}{f(z)}\right)} = \overline{\left(\dfrac{1}{\omega}\right)} = \dfrac{1}{\omega}$

21.

(a) $\zeta = \frac{x+iy}{\frac{1}{2}-z}$

(b) $\zeta' = e^{i\psi}\zeta = \frac{e^{i\frac{\psi}{2}}\zeta + 0}{0\cdot\zeta + e^{-i\frac{\psi}{2}}} = S(\zeta) \qquad S = \begin{pmatrix} e^{i\frac{\psi}{2}} & 0 \\ 0 & e^{-i\frac{\psi}{2}} \end{pmatrix}$

(c)

22.

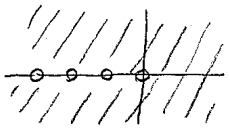
(a) $f_1(z) = \sum_{n=0}^{\infty} z^n = \lim_{n\to\infty} \frac{1-z^n}{1-z} = \frac{1}{1-z}$ $\quad |z|<1$

$f_2(z) = \int_0^{\infty} e^{-(1-z)t}\,dt = \frac{1}{z-1}\left[e^{-(1-z)t}\right]_0^{\infty} = \frac{1}{1-z}$ $\quad \mathrm{Re}\, z < 1$

$f_3(z) = -\frac{1}{z}\sum_{n=0}^{\infty}\frac{1}{z^n} = -\frac{1}{z}\cdot\frac{1}{1-\frac{1}{z}} = \frac{1}{1-z}$ $\quad \left|\frac{1}{z}\right|<1$

(b) $\mathrm{Re}\, z > 0$ $\quad I(z) = \int_0^{\infty} e^{-t}t^{z-1}dt = \left[\frac{1}{z}t^z e^{-t}\right]_0^{\infty} + \frac{1}{z}\int_0^{\infty} e^{-t}t^z dt = \frac{1}{z}I(z+1)$

$I(1) = \int_0^{\infty} e^{-t}dt = 1$

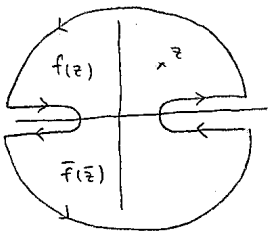


$\mathrm{Re}\, z > -1$ $\quad I(z) = \frac{\Gamma(1+z)}{z} = \frac{1}{z}\int_0^{\infty} e^{-t}t^z dt$

$= \frac{1}{z(z+1)}\Gamma(z+2) = \cdots = \frac{\Gamma(z+n+1)}{z(z+1)\cdots(z+n)}$ $\quad \mathrm{Re}(z+n+1) > 0$

(c) 上半面 z analytic

$\mathrm{Im}\, z > 0$ $\quad \frac{\partial f(z)}{\partial \bar{z}} = 0 \rightarrow \frac{\partial \overline{f(z)}}{\partial z} = \frac{\partial \bar{f}(\bar{z})}{\partial z} = 0$ $\quad \mathrm{Im}\, \bar{z} < 0$



$f(z) = \frac{1}{2\pi i}\oint \frac{f(\zeta)}{\zeta - z}\,d\zeta =$

$= \frac{1}{2\pi i}\int_{-\infty}^{\infty}\frac{f(t)-\bar{f}(t)}{t-z}\,dt$

$= \frac{1}{\pi}\int_{-\infty}^{\infty}\frac{\mathrm{Im}\, f(t)}{t-z}\,dt$

$f(z) = \sum a_n z^n$

$f(\bar{z}) = \sum a_n \bar{z}^n$

$\bar{f}(z) = \sum \bar{a}_n z^n$

$\overline{f(z)} = \sum \bar{a}_n \bar{z}^n$

$\overline{f(\bar{z})} = \sum \bar{a}_n z^n$

23.

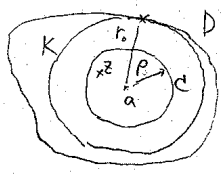
(a) $P(z)$ に零点がない $\rightarrow$ $\frac{P'(z)}{P(z)}$ は pole なし

$0 = \oint_R \frac{P'(z)}{P(z)}\,dz \rightarrow \oint n\frac{1}{z}\,dz = 2\pi n i$ 矛盾

$R\to\infty$

(5) (6)

$f(z)$: 領域 D で正則 $\Rightarrow \forall a \in D$ でベキ級数に展開できる（一様収束）



(Proof)

C : a を中心とした z を含む半径 ρ の円

K : C を含み D に含まれる半径 r_0 の円　$|z-a| < \rho < r_0$

Cauchy の積分定理

$$f(z) = \frac{1}{2\pi i}\oint \frac{f(\zeta)}{\zeta - z}\,d\zeta$$

$$\frac{|z-a|}{|\zeta-a|} = \frac{|z-a|}{\rho} < 1 \qquad |\zeta - a| = \rho$$

$$\frac{1}{\zeta - z} = \frac{1}{(\zeta-a)-(z-a)} = \frac{1}{\zeta-a}\cdot\frac{1}{1-\frac{z-a}{\zeta-a}} = \frac{1}{\zeta-a}\left\{1 + \frac{z-a}{\zeta-a} + \left(\frac{z-a}{\zeta-a}\right)^2 + \cdots\right\}$$

$$\frac{1}{|\zeta-a|}\sum_{n=N}^{\infty}\left|\frac{z-a}{\zeta-a}\right|^n = \frac{1}{\rho}\sum_{n=N}^{\infty}\left(\frac{|z-a|}{\rho}\right)^n \to 0(\zeta) \quad (N\to\infty)$$

一様収束

$$f(z) = \frac{1}{2\pi i}\oint \sum_{n=0}^{\infty}\frac{(z-a)^n}{(\zeta-a)^{n+1}} f(\zeta)\,d\zeta = \sum_{n=0}^{\infty} A_n (z-a)^n \quad :: \quad A_n = \frac{1}{2\pi i}\oint \frac{f(\zeta)}{(\zeta-a)^{n+1}}\,d\zeta$$

A_n の estimation

$\zeta = a + \rho e^{i\theta}$ とおく

$$A_n = \frac{1}{2\pi i}\int_0^{2\pi} i\rho e^{i\theta} d\theta \,\frac{f(a+\rho e^{i\theta})}{(\rho e^{i\theta})^{n+1}} = \frac{1}{2\pi}\frac{1}{\rho^n}\int_0^{2\pi} e^{-in\theta} f(a+\rho e^{i\theta})\,d\theta$$

$$|A_n| \le \frac{1}{\rho^n}\cdot\frac{1}{2\pi}\int_0^{2\pi}|f(a+\rho e^{i\theta})|\,d\theta \le \frac{M(\rho)}{\rho^n}$$

$$M(\rho) = \max_{\theta}|f(a+\rho e^{i\theta})|$$

絶対収束 かつ z について一様収束

ゆえに $\sum_{n=N}^{\infty}|A_n|(z-a)^n \le M(\rho)\sum_{n=N}^{\infty}\left(\frac{|z-a|}{\rho}\right)^n = M(\rho)\left(\frac{|z-a|}{\rho}\right)^N \cdot \frac{1}{1-\frac{|z-a|}{\rho}} \quad (N\to\infty) \to 0$

z の項別微分可能

$$f'(z) = \sum_{n=1}^{\infty} n A_n (z-a)^{n-1}$$

$z = a$; $f'(a) = A_1$

→ 何回でも微分可能

$$A_n = n! f^{(n)}(a)$$

$$\to f^{(n)}(a) = \frac{n!}{2\pi i}\oint \frac{f(\zeta)}{(\zeta-a)^{n+1}}\,d\zeta$$

$f(z)$ ：全複素平面で解析的 ＝ 整関数

1) $f(z)$ const

2) $f(z) = a_0 + a_1 z + a_2 z^2 + \cdots + a_n z^n$

3) $f(z)$ ＝ 無限級数：超越整関数

e^z, $\sin z$, $e^{\frac{1}{z}}$, $\frac{z'}{\sin z}$

平均値の定理 $f(a) = \frac{1}{2\pi}\int_0^{2\pi} f(a + \rho e^{i\theta})\, d\theta$

・Dに含まれる任意の閉領域 $[K]$ で解析関数 $f(z)$ はその最大値を境界でとる.

・さらに $f(z)$ が $[K]$ で $f(z) \neq 0$ とすると $f(z)$ の最小値も境界でとる.

解析関数の零点は孤立している

(Proof) $f(z_0) = 0$

$f(z)$ を z_0 のまわりに展開する

$f(z) = a_0 + a_1(z - z_0) + a_2(z - z_0)^2 + \cdots$

1) $a_0 = 0$, $a_1 = \cdots = a_{n-1} = 0$ $a_n \neq 0$ とすると

$f(z) = (z - z_0)^n (a_n + a_{n+1}(z - z_0) + \cdots)$

$a_n \neq 0$

z_0 の小さい近傍で z_0 以外は 0 にならない.

2) $a_0 = 0$, $a_1 = \cdots$ すべて 0

$f(z_0) = f'(z_0) = f''(z_0) = \cdots = f^{(n)}(z_0) = 0 \rightarrow f(z) = 0$

z_0 のある近傍で恒等的に 0

解析接続

領域（連結開集合）D で $f(z)$, $g(z)$ が解析的かつある点の近傍で $f(z) = g(z)$

$\Rightarrow$ 実は D 全体で $f(z) = g(z)$ ・・・(*)

(*) $\Leftrightarrow$ (*)'

領域 D で $f(z)$ 解析的かつある近傍で $f(z) = 0$ $\Rightarrow$ 実は D 全体で $f(z) = 0$ ・・・(*)'

(Proof) $D' = \{ z \in D \mid f(z) = 0 \} \subset D$

$D' \neq \phi$ ↘ 実は $f'(z) = f''(z) = \cdots = 0$

開集合かつ閉集合

$D^c = D - D'$

$D = D' \cup D^c$

$D' \cap D^c = \phi$

↑ ↗

open

$D^c \neq D$

〈閉集合であること〉

$z_0 \in D' \rightarrow f(z_0) = 0$, $f'(z_0) = 0 \cdots$

そこで $f(z)$ を z_0 のまわりで Taylor 展開する.

z_0 のある近傍で恒等的に 0

D

〈閉集合であること〉

$z_1, z_2 \cdots \in D'$ $z \in D$

$0 = f(z_1) = f'(z_1) = f''(z_1) \cdots$

$= f(z_2) = f'(z_2) = f''(z_2) \cdots$

$f(z)$ の z における連続性から

$0 = f(z) = f'(z) = f''(z) = \cdots$

$f(z)$ はそのある近傍で 0

$z \in D$

(注意) 実は $D' = \{ z \in D, f(z) = 0 \}$ が線分，あるいは集積する点列だけ含んでもよい

(Proof) 集積点 $z_0 \in D'$ で

(注意)

(Proof) 集積点 $z_0 \in D'$ で $f(z)$ を Taylor 展開する.

$f(z) = (z-z_0)^n \{ a_n + a_{n+1}(z-z_0) + \cdots \}$ とかけるが

$f(z_N) = (z_N - z_0)^n \{ a_n + a_{n+1}(z_N - z_0) + \cdots \}$

→ $a_n = 0$ とすると矛盾 → $a_n = 0$ → $a_{n+1} = 0$

つまり D' は集積点のまわりのある近傍をふくむ